

JEE MAIN REVISION MASTER: TRIGONOMETRIC EQUATIONS OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 59 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Trigonometric Functions And Equations — Core Principles & Governing Laws	48 PYQs	Core Principle
High	Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes	11 PYQs	Quantitative Analysis
Standard	Trigonometric Functions And Equations — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Trigonometric Functions And Equations — Core Principles & Governing Laws	$4 \sin 3x = 1, \quad \frac{\cos x}{1+\sin x} = \tan 2x , \quad x \in [0, \frac{\pi}{2}]$
Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes	$[-\frac{\pi}{2}, \frac{\pi}{2}], \quad \sqrt{3} \operatorname{cosec}^2 \theta - 2(\sqrt{3}-1) \operatorname{cosec} \theta - 4 = 0, \quad S = \{x \in [-\pi, \pi] : \sin x (\sin x + \cos x) = a, a \in \blacksquare\}.$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Functions And Equations — Core Principles & Governing Laws (Very High Yield — 48 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Functions And Equations.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Functions And Equations PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "The sum of all the integral values of p such that the equation $3\sin^2x + 12\cos x - 3 = p$, $x \in \blacksquare$, has at least one solution, is:..." [Ans: N/A]
- [2026 Morning Shift] Q#4 (MCQ): "Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0$, $\theta \in [-3\pi, 2\pi]$ is :..." [Ans: N/A]
- [2025 Morning Shift] Q#1 (MCQ): "If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2}\cos^2\theta + (2 - \sqrt{6})\cos\theta - \sqrt{3} = 0$, is equal to:..." [Ans: N/A]

3.2 Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes (High Yield — 11 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\underset{\theta \in S}{\sum} \theta$ is equal to :..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "Let $S = \{x \in [-\pi, \pi] : \sin x(\sin x + \cos x) = a, a \in \blacksquare\}$. Then $n(S)$ is equal to:..." [Ans: D]
- [2025 Evening Shift] Q#1 (MCQ): "The number of solutions of the equation $\cos 2\theta \cos(\theta/2) + \cos(5\theta/2) = 2\cos^3(5\theta/2)$ in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is:..." [Ans: N/A]

3.3 Trigonometric Functions And Equations — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\underset{\theta \in S}{\sum} \theta$ is equal to :..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "The sum of all the integral values of p such that the equation $3\sin^2x + 12\cos x - 3 = p$, $x \in \blacksquare$, has at least one solution, is:..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\sum \theta \in S$ is equal to :

- (A) $-(2\pi / 3)$
- (B) $-(4\pi / 3)$
- (C) $(2\pi / 3)$
- (D) $(4\pi / 3)$

Correct Answer: None

Solution Analysis: Given, $\cos\theta + 1 = \sqrt{3}\sin\theta$ We have to find all $\theta \in (-2\pi, 2\pi)$ satisfying this equation, and then calculate their sum. Rewrite the equation as $1 + \cos\theta = \sqrt{3}\sin\theta$ Now use the standard identities: $1 + \cos\theta = 2\cos^2(\theta / 2)$ and $\sin\theta = 2\sin(\theta / 2)\cos(\theta / 2)$ Substituting these, $2\cos^2(\theta / 2) = \sqrt{3} \cdot 2\sin(\theta / 2)\cos(\theta / 2)$ So, $2\cos(\theta / 2)(\cos(\theta / 2) - \sqrt{3}\sin(\theta / 2)) = 0$ Hence, either Case 1: $\cos(\theta / 2) = 0$ So, $(\theta / 2) = (\pi / 2) + n\pi$ which gives

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#2)

The sum of all the integral values of p such that the equation $3\sin^2x + 12\cos x - 3 = p$, $x \in \mathbb{R}$, has at least one solution, is:

- (A) -54
- (B) -60
- (C) -75
- (D) -84

Correct Answer: None

Solution Analysis: Given $3\sin^2x + 12\cos x - 3 = p$, $x \in \mathbb{R}$ We need all integral values of p for which this equation has at least one real solution. Now use the identity $\sin^2x = 1 - \cos^2x$ So, $3(1 - \cos^2x) + 12\cos x - 3 = p$ Simplifying, $3 - 3\cos^2x + 12\cos x - 3 = p$ $-3\cos^2x + 12\cos x = p$ Let $t = \cos x$ Then $-1 \leq t \leq 1$ So we have $p = -3t^2 + 12t$ Now we need the range of $f(t) = -3t^2 + 12t$, $-1 \leq t \leq 1$ This is a downward opening parabola. Find its values at the endpoints: $f(-1) = -3(1) + 12(-1) = -3 - 12 = -15$ $f(1) = -3(1) + 12(1) = -3 + 12 = 9$ Also, $f'(t) = -6t + 12$ For $-1 \leq t \leq 1$, we have $-6t + 12 > 0$ so $f(t)$ is increasing